Where to Hitch? Finding Small, Rare Parts Across Long Spatiotemporal Contexts

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Abstract

Foundation models struggle to find small, rare parts with few distinctive visual features in long, high-resolution videos. Resizing frames hides small details, while searching every frame in detail is costly. We study hitch assist: locating a trailer's coupler, the socket that receives the vehicle's hitch ball, as the vehicle backs toward it. In our densely annotated videos, the coupler covers a median 0.2% of the image, and text-prompted models often box a larger structure. We introduce **HitchIt**, which chooses frames and image regions for a pretrained video model to search, without further training. Boxes in reference images show the model what to find. HitchIt searches a few late frames, when the coupler appears larger, narrowing the search from the trailer through its projecting frame, or tongue, to the coupler. Cropping before resizing enlarges the coupler in the model's input image and excludes other objects. A selected box then initializes tracking backward and forward through the video. On 20 densely annotated videos, HitchIt reaches 0.627 mean box overlap (mIoU), versus at most 0.067 for text-prompted models, including SAM3 on the same searched frames. On 67 separate videos, evaluated on one labeled frame each, part-by-part search raises mIoU from 0.513 to 0.645 over full-image search. We will release our dense coupler annotations.

1 Introduction

Hitch-assist systems must locate a trailer's coupler as the vehicle backs toward it [1, 2]. The coupler is the socket that receives the vehicle's hitch ball. It sits at the tip of the trailer's tongue, the frame that projects toward the vehicle (Fig. 2a). It is small: in a rear-camera video, it appears tiny at first and larger as the vehicle approaches (Fig. 1a). It is rare: a specialized towing part, uncommon in everyday images. It is visually plain: similar metal surfaces appear on many objects, including nearby jacks, chains, other trailers, and the vehicle's hitch ball, so appearance alone does not distinguish it.

The coupler's size makes its localization a long-context problem in both space and time. A recording spans many high-resolution frames, but a foundation model sees each input at a fixed resolution. Processing every frame in full detail is costly, and downsizing the frames hides the part. The coupler's rarity and plain appearance add a second difficulty: its name does not identify it. Text-prompted foundation models such as Grounding DINO [3], OWLv2 [4], and SAM3 [5] often return a box around a larger structure, such as the trailer, instead of the coupler.

A reliable detection in a clear frame can initialize tracking through the rest of the video. The search can therefore concentrate on frames where the coupler is easier to see. Later frames show the coupler at a larger scale, but the final frame need not provide the best starting box (Fig. 1a). Each search region must retain enough detail to distinguish the coupler from other objects.

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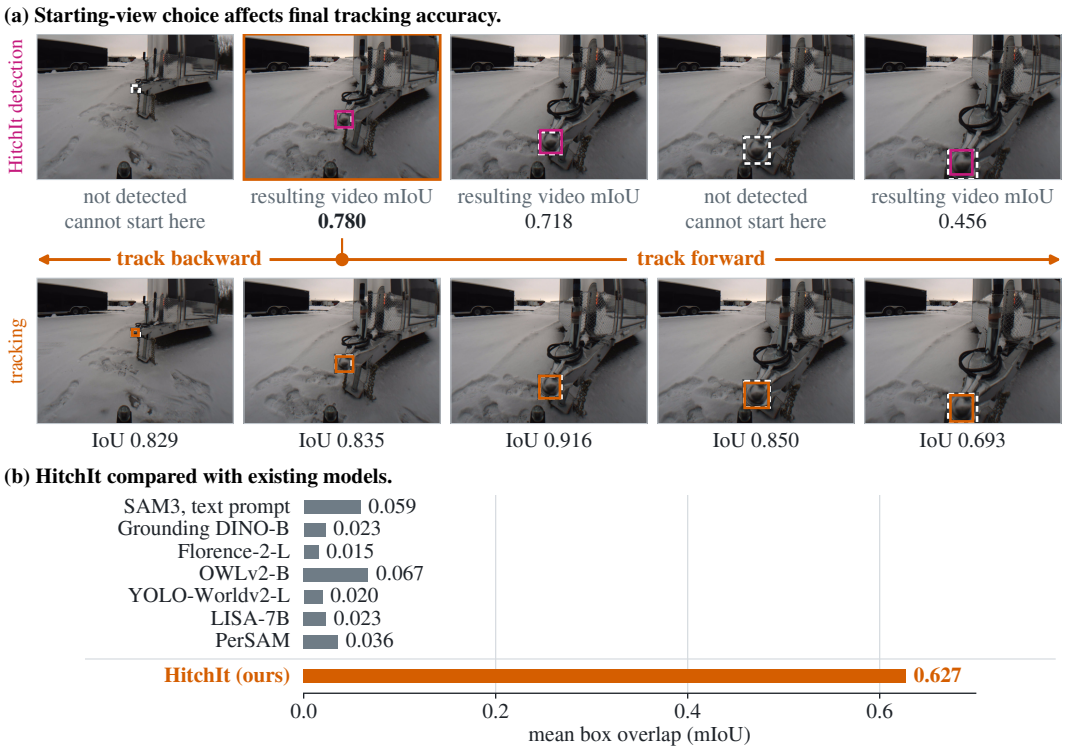


Figure 1: **Finding a suitable starting view improves the resulting track.** **(a)** Top: HitchIt's detection, with retrieved examples and part-by-part search, runs on each view separately. A detected box starts the same tracker, and the resulting video mIoU is measured on the same video. A view without a detection cannot start a track. Bottom: boxes tracked from the highlighted detection, with per-frame IoU. Magenta: HitchIt detections; orange: tracked boxes; dashed white: annotations. **(b)** Published baselines [3–9] and HitchIt on 20 videos; SAM3 uses text prompts on four late views.

We introduce HitchIt, which finds this starting box automatically using a pretrained video model without further training (Fig. 3). HitchIt retrieves reference images from other videos, each with a box marking the target part, and chooses them by the part's appearance. It places each example beside the search image as one input, so the reference box specifies the target without an annotation in the new video. The search proceeds from the trailer to its tongue to the coupler. The trailer and tongue are larger and easier to find than the coupler. Cropping around them before resizing makes the coupler larger in the model's input image and leaves out objects elsewhere. HitchIt searches four late views and tracks the selected box backward and forward through the video.

We evaluate HitchIt on 20 densely annotated approach videos and on 67 further videos with one labeled frame each. On the dense videos, HitchIt localizes the coupler far more accurately than text-prompted foundation models (Fig. 1b). With the same model and late views, text prompts still perform poorly, so size alone does not explain the difficulty. Identification succeeds mainly in late views, where the coupler is large. Part-by-part search outperforms fixed image tiles, and on the 67 separate videos it clearly improves on searching the full image. These results support identifying the target in a few clear views and focused image regions, then tracking it through the recording.

Our contributions are as follows. (1) **Selective context for hard-to-find parts:** HitchIt identifies a small, rare, and visually plain part by choosing which views, image regions, and examples a pretrained model sees, without further training or a target box in the new video. (2) **Controlled evidence:** holding the search or the tracker fixed, we measure how view choice, example retrieval, and part-by-part search each affect localization. (3) **Dense coupler annotations:** we annotate coupler boxes throughout 20 approach videos from the MSU Trailer Database [1] and will release them.

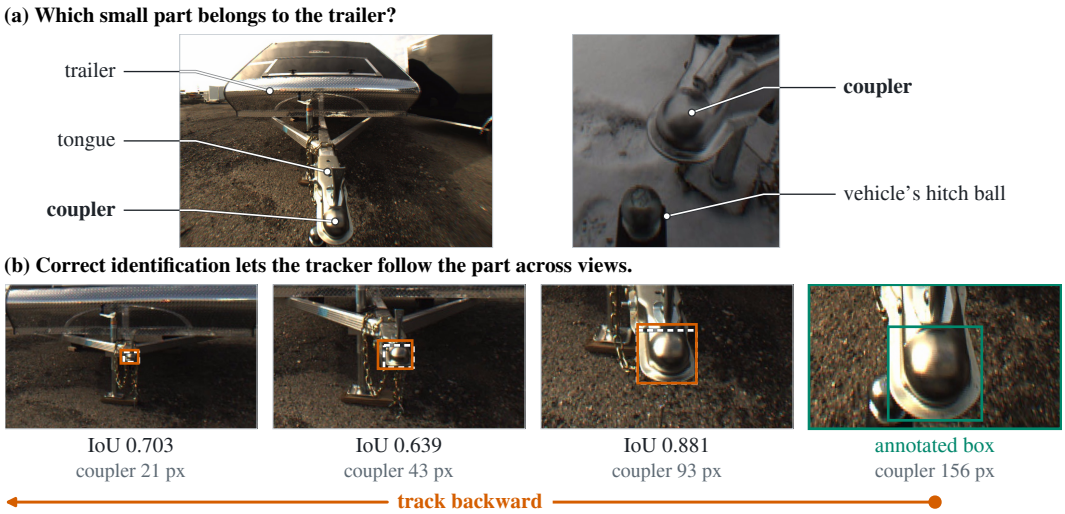


Figure 2: **The coupler is difficult to identify, but a correct starting box supports tracking. (a)** The coupler sits at the tip of the trailer's tongue; the nearby hitch ball belongs to the vehicle. **(b)** The tracker receives the annotated coupler box in the rightmost view (green) and predicts boxes in earlier views (orange). Dashed white: annotations used only for evaluation. Close-ups share one display scale; coupler size is $\sqrt{\text{box area}}$ in original pixels. Starting from an annotated box tests tracking separately from detection; automatic localization must also find the coupler in a suitable frame.

2 Related Work

Long videos and memory-based tracking. Long-video benchmarks test whether models can use information spread across a recording [10–12]. Memory-based segmenters follow a target through a video once it is marked in one frame [5, 13–15]. HitchIt uses such a tracker unchanged and addresses the step before it: finding the target automatically in a few chosen views. Selective labeling picks samples for human annotation [16, 17]; HitchIt picks views for automatic localization.

Text and visual prompts. Open-vocabulary detectors and segmenters localize regions named by text [3, 4, 6–8]. Visual prompting instead specifies the target with examples [5, 18–20]. Training-free methods segment new images from annotated references using frozen features [9, 21, 22]. HitchIt also uses examples without training. It retrieves examples by part appearance from other videos and places each beside the search image, so a reference box can guide detection in a new image.

Guided visual search. V* [23] lets a multimodal language model search a high-resolution image for small details by zooming into likely regions. HitchIt also zooms selectively, but it uses larger surrounding structures to guide the search, from the trailer to its tongue and then to the coupler.

Coupler perception. Prior hitch-assist systems train detectors on task-specific labels [1, 2]. The MSU Trailer Database provides approach videos with coupler centers and close-up contours [1]. We study training-free 2D localization and add dense coupler boxes for 20 of its videos.

3 HitchIt

Task. HitchIt receives an unannotated approach video and a bank of boxed examples of the trailer, tongue, and coupler from other videos. It predicts coupler boxes throughout the video. Offline processing lets HitchIt use a clear late view to localize the coupler in earlier frames.

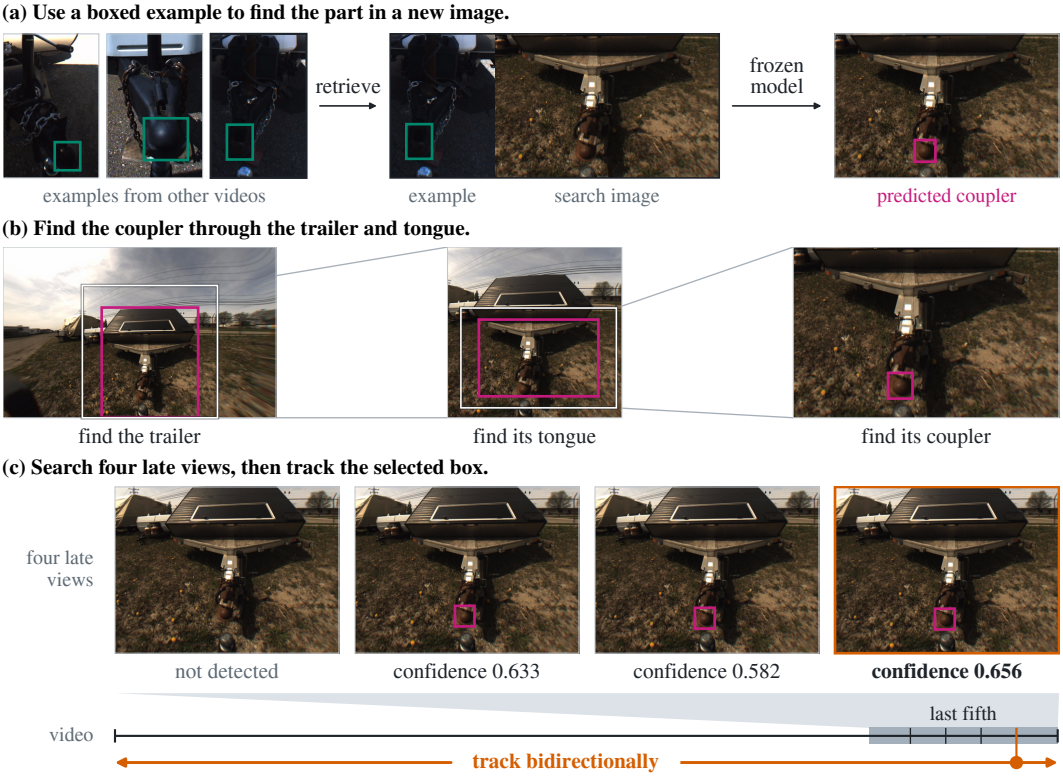


Figure 3: **HitchIt chooses the examples, image regions, and views that the model sees.** **(a)** A boxed example from another video is paired with the unmarked search image, here at the coupler step; the prediction comes from that pair. **(b)** Predicted trailer and tongue boxes narrow the search to the coupler, each step using (a); white outlines mark the region searched at the next step. **(c)** The search in (b) runs on four views from the last fifth of the video, and the selected box is tracked to both ends. Labels give detection confidence, not tracking accuracy. Magenta: detections; green: annotated examples; orange: tracking. These choices focus the search on a clear view of the part.

Overview. HitchIt uses a pretrained video model without further training, relying on two abilities. Given a box around an example object, the model finds matching objects in the same input image. Given a box in one frame, it tracks the object through the video. The model resizes every input to a fixed resolution, which limits the detail it sees. HitchIt decides how to use this fixed input (Fig. 3): which examples to show, which image region to search, and which views to search before tracking.

3.1 Specify the part with examples from other videos

The coupler is a rare concept, and a name such as "trailer coupler" does not reliably identify it. HitchIt therefore specifies the part with boxed examples. For each part, the bank holds examples with varied appearances, viewpoints, and backgrounds, all from videos other than the one being processed.

Retrieve examples by part appearance. A useful example resembles the target part, not merely the scene around it. HitchIt therefore ranks examples by *part matching*. It averages the model's image features inside each example's box and compares the resulting vector with the features at each location in the search image. An example's score is the mean of its eight highest cosine similarities, and HitchIt keeps the eight best-scoring examples.

Pair each example with the search image. The model's visual prompt expects the example box and the target in the same image. HitchIt therefore places a retrieved example beside the search image and gives the pair to the model as one input, with a box only on the example (Fig. 3a). It keeps the most confident predicted box on the search-image side. The example's box thus specifies what to

find, not where to find it. Each of the eight examples forms its own pair. HitchIt removes duplicate overlapping detections with non-maximum suppression and ranks the boxes by model confidence.

3.2 Search from the trailer to its coupler

In a full-image pair, the search image occupies half of the model's fixed input, leaving few pixels for a small coupler. The tongue extends from the trailer body, and the coupler sits at its tip. The trailer and tongue are larger and easier to find, so they can guide the search to the coupler. HitchIt therefore narrows the search through them (Fig. 3b). It first finds the trailer in the full image, then the tongue inside a crop around each trailer, and then the coupler inside a crop around the tongue. Each stage pairs its crop with examples of that part; coupler examples show a boxed coupler within a crop of its tongue. Each crop enlarges the next part in the resized input image and excludes objects elsewhere.

HitchIt keeps up to three candidate trailers and rejects tongues and couplers with implausible size or position. For example, a tongue must lie in the lower part of its trailer box, and a coupler box may cover at most a quarter of its tongue box's area. When the tongue crop yields no valid coupler, HitchIt searches the full image for a coupler inside the detected tongue, and then searches inside the trailer.

3.3 Identify once, then track

Searching N frames with all R reference examples takes NR model calls per part. With 99 examples per bank, this would require $4{,}851$ to $17{,}424$ calls per part per video. HitchIt instead searches four evenly spaced frames in the last fifth of the video, where the approaching camera sees the coupler largest (Fig. 3c). Among the valid coupler boxes from these views, it prefers boxes found through a detected tongue over boxes found only in the trailer. It then selects the most confident box, breaking near-ties toward the larger box, and slightly enlarges the box if it is small. The selected box starts the model's tracker on its frame, and tracking runs to both ends of the video (Fig. 1a). If no view yields a valid coupler, HitchIt outputs no track rather than following a larger part.

4 Experiments

We first compare HitchIt with existing methods. We then isolate the effects of view choice, example retrieval, and part-by-part search, and finally test the complete method on separate videos.

4.1 Experimental setup

Data. We use rear-camera approach videos from the MSU Trailer Database [1]. Our densely annotated set has 20 videos with 49–176 sampled 1920×1200 frames each, $1{,}784$ frames in total. We annotated coupler boxes on $1{,}136$ of these frames and score all methods on them. The remaining frames, all at the start of a video, carry no annotation and are not scored. The coupler covers a median 0.2% of the image, and within a video its size ($\sqrt{\text{box area}}$) varies by a median factor of 6.6 between the 10th and 90th percentiles. Each example bank holds 99 boxed examples from 99 videos outside this set. HitchIt has one bank per part, and its coupler examples are crops around the tongue.

Metrics. Box intersection over union (IoU) divides intersection area by union area: 0 means no overlap and 1 an exact match. Mean IoU (mIoU) averages the IoU of the best-matching predicted box over annotated frames, counting a missing prediction as zero. AP_{50} and AR_{50} are COCO average precision and recall [24] at IoU 0.5; recall allows one box per frame. We adapt cumulative IoU (cIoU) from referring segmentation [8] to boxes. It divides the total intersection by the total union over annotated frames, including unmatched predicted boxes in the denominator, so oversized and extra boxes lower it. mIoU is our main metric because it credits partial overlap without relying on confidence scores, which are constant within each track.

Implementation. We build HitchIt on SAM3 [5] and use its exemplar prompts and box-initialized video tracker unchanged. Part matching uses features from SAM3's frozen Perception Encoder [25]. SAM3 resizes each input to 1008×1008, so in a full-image pair the search image receives 504 pixels of input width. We convert the tracker's masks to tight boxes.

Table 1: **HitchIt localizes the coupler more accurately than published baselines.** Results on the 1,136 annotated frames of the 20 densely annotated videos. Tracked rows search the listed views, then track. Text*: the better of one or three target names. Bold: best automatic result. The annotated-box row measures tracking from a supplied correct box rather than automatic localization.

Method	Prompt	Searched views	mIoU	AP_{50}	AR_{50}	cIoU
Text-prompted foundation models						
SAM3 [5], first frame	text	first frame → track	0.038	0.000	0.000	0.106
SAM3 [5], late views	text	4 late views → track	0.059	0.000	0.000	0.099
Grounding DINO-B [3]	text	every frame	0.023	0.000	0.000	0.022
Florence-2-L [6]	text	every frame	0.015	0.000	0.000	0.016
OWLv2-B [4]	text*	every frame	0.067	0.000	0.002	0.097
YOLO-Worldv2-L [7]	text*	every frame	0.020	0.000	0.000	0.013
LISA-7B [8]	text	every frame	0.023	0.000	0.000	0.020
Example-prompted methods: no box in the new video						
PerSAM [9]	boxed examples	4 late views → track	0.036	0.000	0.003	0.014
HitchIt (ours)	boxed examples	4 late views → track	**0.627**	**0.564**	**0.750**	**0.632**
Oracle reference: annotated box supplied in the test video						
Annotated box	test-video box	1 frame → track	0.683	0.616	0.787	0.759

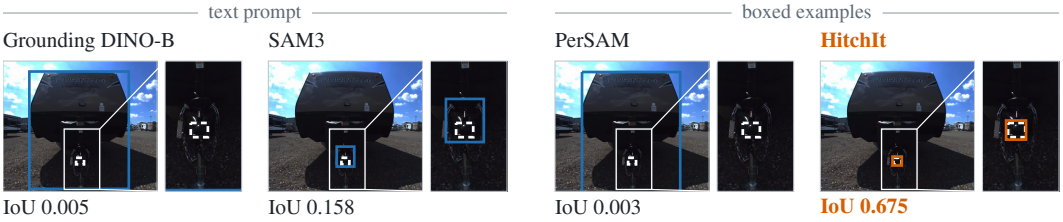


Figure 4: **Finding a larger structure does not identify the coupler.** Grounding DINO-B [3], late-view SAM3 text prompting [5], PerSAM [9], and HitchIt on the same frame, each with a full view and a close-up of the outlined region. PerSAM follows Table 1's boxed-example and tracking protocol; its box extends below the full view. Blue: baseline predictions; orange: HitchIt; dashed white: the annotated coupler box. Only HitchIt isolates the coupler in this example.

Compared methods. We use "trailer coupler" as the default text prompt. Grounding DINO-B [3], Florence-2-L [6], OWLv2-B [4], YOLO-Worldv2-L [7], and LISA-7B [8] process every frame independently and return one box per frame. SAM3 receives the name either as a video prompt on the first frame or as an image prompt on four late views, whose most confident box starts SAM3's tracker. PerSAM [9] receives eight full-image coupler examples per view, retrieved by part matching, with their boxes converted to masks. It uses HitchIt's four late views and tracker. For an oracle reference, we start the tracker from each video's last annotated box (Fig. 2b).

Ablation baseline. *Direct search* skips the trailer and tongue stages. It retrieves full-image coupler examples by part matching and searches the full image, with HitchIt's views and tracker.

4.2 Comparison with existing methods

HitchIt localizes the coupler more accurately than text prompting. In Table 1, HitchIt reaches 0.627 mIoU. Text-prompted settings reach 0.015 to 0.067, all with an AP_{50} of 0.000. Even on the same four late views, SAM3 reaches only 0.059. Size alone does not explain the gap: for couplers of at least 80 pixels (517 frames), text prompts reach at most 0.108 mIoU, versus 0.715 for HitchIt.

Baselines can box the wrong structure. Figure 4 shows Grounding DINO-B and PerSAM boxing the trailer. Late-view SAM3 text prompting boxes the hitch assembly, while HitchIt boxes the coupler. PerSAM fails even though it also receives boxed coupler examples from other videos: it reaches only 0.036 mIoU. On 92.5% of frames, Grounding DINO-B returns a box that covers at least half of the coupler but is at least four times its area; its median box is 63 times the coupler's area.

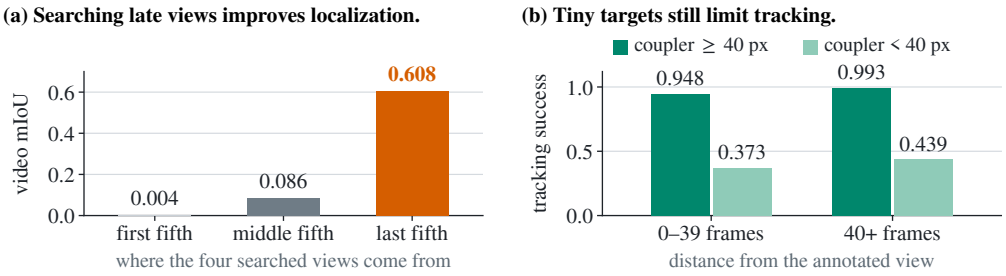


Figure 5: **Late views aid identification; tiny couplers can still limit tracking. (a)** Direct search on four views from the first, middle, or last fifth of each video, followed by the same tracker. **(b)** With an annotated starting box, the fraction of frames tracked correctly (box IoU ≥ 0.5), grouped by distance from that view in sampled frames and by coupler size.

Matching size is not enough to identify the coupler. On 752 of the 1,136 frames, OWLv2 proposes a box overlapping the coupler with IoU ≥ 0.5, but its text score selects such a box on only two. A detailed description does not help either. With the prompt "small metal socket at the tip of the trailer tongue", Grounding DINO-B's median predicted area shrinks to 1.6 times the coupler's area, but only two frames reach IoU ≥ 0.5. Its errors include a frame fitting, a decal, and the vehicle's hitch ball, so matching the coupler's scale does not single it out.

4.3 Temporal context: search late views, then track

Late views make the coupler findable. To isolate the choice of views, we fix direct search and the tracker and change only where the four searched views come from: the first, middle, or last fifth of each video. If none of the four views yields a box, no track is produced. In Fig. 5a, mIoU rises from 0.004 for the first fifth and 0.086 for the middle fifth to 0.608 for the last fifth. The coupler is largest near the end of an approach, so late views give the search the most pixels on the part. The final view need not be the best starting point. In the controlled example in Fig. 1a, tracking from the highlighted detection gives 0.780 mIoU, versus 0.456 when starting from the final view.

A correct starting box enables effective tracking. Figure 2b illustrates how one correct box lets the tracker follow the coupler through earlier views. In Table 1, the oracle reference, which starts the same tracker from the annotated box, gives 0.683 mIoU across all 20 videos, whereas the name prompt on the late views gives 0.059. These results show the value of a correct starting box in a suitable frame: HitchIt reaches 0.627 with an automatically selected box.

Tiny couplers remain hard to track. Figure 5b groups tracking success (IoU ≥ 0.5) from the annotated starting box by distance from that view and by coupler size. For couplers at least 40 pixels in size ($\sqrt{\text{box area}}$), success is 0.948 within 39 sampled frames of the starting view and 0.993 beyond. For smaller couplers, the corresponding success rates are 0.373 and 0.439.

4.4 Spatial context: where the gains come from

Table 2 varies how examples are retrieved and where the coupler is searched, with the four late views and the tracker fixed. *Scene matching* ranks examples by the similarity of whole-image embeddings, from CLIP [26] or from SAM3's encoder, instead of part matching. Two spatial baselines use HitchIt's cropped coupler examples without trailer or tongue predictions. One searches the full image, and the other searches six fixed, overlapping tiles, each half the image width and two-thirds its height.

Match the part, not the scene. Ranking examples by the part's appearance gives the best direct-search mIoU (0.608), above whole-scene ranking with CLIP (0.590) or with SAM3 features (0.544).

Part-by-part search beats fixed crops. With the same cropped coupler examples, part-by-part search reaches 0.627 mIoU, versus 0.460 for six fixed tiles and 0.486 for full-image search. The tiles require six times as many model calls as full-image search. Part-by-part search also improves mIoU, AP_{50}, and AR_{50} over direct search with full-image examples.

Table 2: **Part matching and part-by-part search improve localization.** All rows use the same four late views of the 20 densely annotated videos and the same tracker. The first group varies example retrieval; the last two compare search regions with cropped coupler examples. Bold: best.

Search setup	mIoU	AP_{50}	AR_{50}	cIoU
Direct search: how examples are retrieved				
Scene matching (CLIP [26])	0.590	0.456	0.676	0.587
Scene matching (SAM3 features)	0.544	0.413	0.634	0.542
Part matching (SAM3 features)	0.608	0.512	0.705	**0.657**
Direct search with cropped coupler examples				
Full image	0.486	0.442	0.607	0.485
Six fixed tiles	0.460	0.360	0.555	0.465
Part-by-part search				
HitchIt: trailer $\rightarrow$ tongue $\rightarrow$ coupler	**0.627**	**0.564**	**0.750**	0.632

Table 3: **Part-by-part search improves on direct search across separate videos.** Results on 67 videos, each scored on one labeled late frame using the tracked box. Every test video is excluded from its example bank. Both methods use part matching and the same tracker. This tests cross-video localization, not tracking accuracy across all frames.

Search setup	mIoU	AP_{50}	AR_{50}	cIoU
Direct search	0.513	0.349	0.537	0.438
HitchIt (part-by-part search)	**0.645**	**0.718**	**0.761**	**0.649**

4.5 Evaluation on a wider range of videos

We evaluate HitchIt on 67 further videos from the MSU Trailer Database. Each has one annotated coupler box on a frame 90 to 93% through the video. We split the videos into five groups and process each group with example banks built from the other four groups, which gives 53 or 54 examples per part. We use each held-out video's annotation only to score its tracked box.

Part-by-part search holds up on separate videos. In Table 3, part-by-part search improves all four metrics over direct search. It raises mIoU from 0.513 to 0.645 and roughly doubles AP_{50}, from 0.349 to 0.718. The mIoU gain is positive in all five video groups, from 0.003 to 0.244.

5 Conclusion

We introduced HitchIt for locating a trailer's coupler, the socket that receives a vehicle's hitch ball, throughout a recorded approach video. HitchIt uses a pretrained video model without further training. Reference images with boxes marking the target part show the model what to find. HitchIt searches a few frames near the end of the approach, when the coupler appears larger. In each searched frame, it narrows the search from the trailer to its projecting frame, or tongue, and finally to the coupler. Cropping before resizing makes the coupler larger in the model's input image and leaves out objects that could be mistaken for it. Tracking runs backward and forward from the selected box.

On 20 densely annotated videos, HitchIt achieves a mean box overlap (mIoU) of 0.627, compared with at most 0.067 for the evaluated text-prompted models and PerSAM. On 67 additional videos, evaluated on one labeled frame per video, searching through the trailer and tongue improves mIoU from 0.513 to 0.645 over direct full-image search. These results support identifying the part in a suitable view before tracking it through the recording, although very small couplers remain difficult to track. We will release our dense coupler annotations. The same approach may help locate small, uncommon parts in other recorded videos where nearby objects look similar.

Acknowledgments

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